

Multiple Choice (10 marks)

1. All of the following are examples of hydrolysis EXCEPT
 - (a) conversion of fats to fatty acids and glycerol
 - (b) conversion of proteins to amino acids
 - (c) conversion of starch to simple sugars
 - (d) conversion of pyruvic acid to acetyl CoA
2. Keystone species are thought to have profound effects on the structure and composition of ecological communities because they
 - (a) are more abundant than most other species in their communities
 - (b) tend to reduce diversity by eliminating food resources for other species
 - (c) are the largest species in their communities
 - (d) are the most aggressive species in their communities
 - (e) are the only species capable of reproduction
3. A feature of amino acids NOT found in carbohydrates is the presence of
 - (a) carbon atoms
 - (b) oxygen atoms
 - (c) nitrogen atoms
 - (d) hydrogen atoms
4. Which of the following plant cells undergoes programmed cell death to become functional?
 - (a) Epidermal cell
 - (b) Stomatal guard cell
 - (c) Root cap cell
 - (d) Xylem vessel member
 - (e) Phloem sieve tube member
5. A deck of playing cards is thoroughly shuffled and a card is drawn from the deck. What is the probability that the card drawn is the ace of diamonds?
 - (a) $1/53$
 - (b) $3/52$
 - (c) $2/51$
 - (d) $2/52$
 - (e) $4/52$

True / False (10 marks)

Answer True or False

6. True or False: Cardiopulmonary bypass temperature does not affect postoperative euthyroid sick syndrome?
7. True or False: Ultrasound in squamous cell carcinoma of the penis; a useful addition to clinical staging?
8. True or False: Factors determining the survival of nasopharyngeal carcinoma with lung metastasis alone: does combined modality treatment benefit?
9. True or False: Does base deficit predict mortality in patients with severe traumatic brain injury?
10. True or False: Can the condition of the cell microenvironment of mediastinal lymph nodes help predict the risk of metastases in non-small cell lung cancer?

Long Form Response (20 marks)

Answer each question with a clear and complete explanation.

11. Discuss the distinctions between innate and adaptive immunity, focusing on their mechanisms and roles in the immune response, and identify common misconceptions related to these systems.
12. Discuss the various functions of bones beyond locomotion and support, emphasizing their role in digestion and other physiological processes. Analyze how these functions contribute to overall health.

End of Paper